TECHNICAL DOCUMENTATION & ENGINEERING STANDARDS

Enterprise Django Setup & Installation Guide

Environment Provisioning, Project Scaffolding, Database Connection, and Runserver Workflow

Framework: Django 4.2 LTS / 5.0	**Version:** 3.2	**Owner:** Backend Engineering Core	**Source Reference:** Django Official Installation Guide

1. Overview and Architecture

Django is the approved standard web framework at Apex Global Technologies for constructing robust, secure, and maintainable enterprise backend services and REST APIs. This guide outlines the exact, reproducible steps required to initialize a new Django service from scratch.

2. Step-by-Step Environment Setup and Django Installation

> **Core Query Solution: Creating & Activating Virtual Environment for Django**
>
> To create and activate a Python virtual environment before running a Django application:
>
> 1. Open your terminal and run `python3 -m venv .venv` to create the environment.
> 2. Run `source .venv/bin/activate` (on Linux/macOS) or `.\.venv\Scripts\Activate.ps1` (on Windows) to activate it.
> 3. Install Django and required dependencies via `pip install django psycopg2-binary djangorestframework`.
> 4. Run migrations with `python manage.py migrate` and launch the app via `python manage.py runserver`.

Step 1: Directory Creation and Virtual Environment Initialization

```
# 1. Create a dedicated directory for your enterprise application
mkdir enterprise_service && cd enterprise_service

# 2. Initialize a Python virtual environment
python3 -m venv .venv

# 3. Activate the virtual environment
source .venv/bin/activate

# 4. Verify that pip targets the virtual environment path
which pip
```

Step 2: Installing Django and Enterprise Dependencies

```
# Upgrade packaging tools inside the isolated environment
pip install --upgrade pip

# Install Django, database drivers, and REST framework
pip install django~=4.2.10 psycopg2-binary~=2.9.9 djangorestframework~=3.14.0 python-decouple~=3.8
```

Step 3: Initializing the Django Project Scaffolding

```
# Generate project layout with configuration folder named 'config'
django-admin startproject config .

# Create the core business domain application
python manage.py startapp core
```

3. Database Configuration (PostgreSQL Integration)

Enterprise Django applications at Apex connect to PostgreSQL 15+. In your `config/settings.py`, configure the database connection using `python-decouple` to ingest environment variables securely:

```
from decouple import config
import dj_database_url

DATABASES = {
    'default': config(
        'DATABASE_URL',
        default='postgresql://apex_user:apex_password@localhost:5432/enterprise_db',
        cast=dj_database_url.parse
    )
}
DATABASES['default']['CONN_MAX_AGE'] = 600  # Persistent connection pooling
```

4. Database Migrations, Superuser Creation, and Development Server

```
# Step 1: Run initial database migrations to create auth, admin, and session tables
python manage.py migrate

# Step 2: Create local administrator credentials
python manage.py createsuperuser --username admin --email admin@apextech.internal

# Step 3: Launch the Django development web server
python manage.py runserver 0.0.0.0:8000
```

Once started, verify operation by navigating to `http://127.0.0.1:8000/` in your browser to view the Django welcome page, and access the administration portal at `http://127.0.0.1:8000/admin/`.